GPSC CBRT · HOMOEOPATHIC PHYSICIAN · CORE

Topic 9: Surgery, Allied Specialties & First-Aid

Complete notes with pictures, memory hooks, rapid revision and practice MCQs.

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Topic 9: Surgery, Allied Specialties & First-Aid

How to use this guide

Topic 9 gives about 2 of the 25 core marks. Questions ask a definition (sinus or fistula), a classic sign (Colles' fracture, halos round lights), a first-aid step, or the homoeopathic remedy for a type of injury.

Syllabus heading (Advt 06/2026)	What it covers	Section
General surgical conditions	Wounds, abscesses, shock, ulcers, burns, perioperative care	9.1
Traumatology and orthopaedics	Fractures, dislocations, sprains, osteomyelitis, osteoarthritis, cervical and lumbar spondylosis	9.2
ENT and ophthalmology	Sinusitis, otitis media, tonsillitis, conjunctivitis, cataract, glaucoma	9.3
Surgical therapeutics and first aid	Homoeopathic remedies for mechanical injuries, pre- and post-operative care, snakebite and poisoning	9.4

- **Standard books.** S. Das, *A Concise Textbook of Surgery* and *A Manual on Clinical Surgery*; *Bailey & Love's Short Practice of Surgery*; Maheshwari, *Essential Orthopaedics*; P. L. Dhingra, *Diseases of Ear, Nose and Throat*; A. K. Khurana, *Comprehensive Ophthalmology*; St John Ambulance / Indian Red Cross first-aid manual; national snakebite protocol (MoHFW).
- Homoeopathic surgical therapeutics: Boericke's *Materia Medica*; Allen's *Keynotes*; E. B. Nash, *Leaders in Homoeopathic Therapeutics*; Shepherd and Lilienthal for injuries.
- Drug treatments from conventional pharmacology are left out; first aid and surgical principles are kept because the syllabus asks for them.

9.1 General surgical conditions

Definitions carry most marks here: the types of wound healing, sinus versus fistula, the kinds of shock and ulcer, and burn depth with the rule of nines.

Wounds and healing

Term	Meaning
Healing by first intention (primary)	Clean incised wound, edges brought together; minimal scar (surgical incision)

Term	Meaning
Healing by second intention (secondary)	Wound left open or infected; fills with granulation tissue from the base; wide scar, contraction
Third intention (delayed primary closure)	Contaminated wound cleaned and closed after 3–5 days
Phases of healing	Haemostasis → inflammation (days 1–4) → proliferation, granulation (day 4 to 3 weeks) → remodelling (up to a year or more)
Types of wound	Incised (clean edges, bleeds freely), lacerated (torn, irregular), abrasion (graze), contusion (bruise, skin intact), punctured (deep, risk of tetanus)
Keloid	Scar growing **beyond** the wound margins, recurs after excision; hypertrophic scar stays **within** the margins
Wound dehiscence	Bursting open of a surgical wound (burst abdomen), often days 5–8

Factors delaying healing: infection (commonest), poor blood supply, diabetes, malnutrition (protein, vitamin C, zinc), steroids, foreign body, old age.

Abscess, cellulitis, sinus and fistula

Term	Definition
Abscess	Localised collection of pus in a cavity lined by granulation tissue (pyogenic membrane); commonest organism *Staphylococcus aureus*; "where there is pus, let it out" (incision and drainage)
Cellulitis	Diffuse spreading infection of skin and subcutaneous tissue; commonest organism *Streptococcus pyogenes*
Sinus	A blind track lined by granulation tissue, opening onto an epithelial surface (e.g. pilonidal sinus)
Fistula	An abnormal communication between **two epithelial surfaces** (e.g. fistula-in-ano)
Carbuncle	Infective necrosis of several adjacent hair follicles; back of the neck in diabetics
Furuncle (boil)	Infection of a single hair follicle

Hook: "A Sinus is a Single opening; a Fistula is a Further connection."

Shock

Shock is inadequate tissue perfusion. The commonest type after injury is **hypovolaemic**.

Type	Cause	Signs
Hypovolaemic	Blood or fluid loss (haemorrhage, burns, vomiting, diarrhoea)	Cold clammy skin, rapid thready pulse, low BP, low urine output
Cardiogenic	Pump failure (myocardial infarction)	Cold skin, raised JVP
Septic (distributive)	Infection releasing toxins	Early warm flushed skin and bounding pulse, later cold
Anaphylactic (distributive)	Allergy	Urticaria, stridor, wheeze, low BP
Neurogenic (distributive)	Spinal cord injury	Low BP with **slow** pulse, warm skin
Obstructive	Tension pneumothorax, cardiac tamponade	Raised JVP, low BP

- **Blood loss and signs (ATLS classes)**: up to 15% (750 mL) little change; 15–30% pulse over 100; 30–40% BP falls; over 40% life-threatening.
- Urine output under 0.5 mL/kg/h signals poor perfusion.
- **Hook for shock first aid: lie flat, legs raised, keep warm, nothing by mouth, shift to hospital.**

Ulcers

An ulcer is a break in the continuity of an epithelial surface.

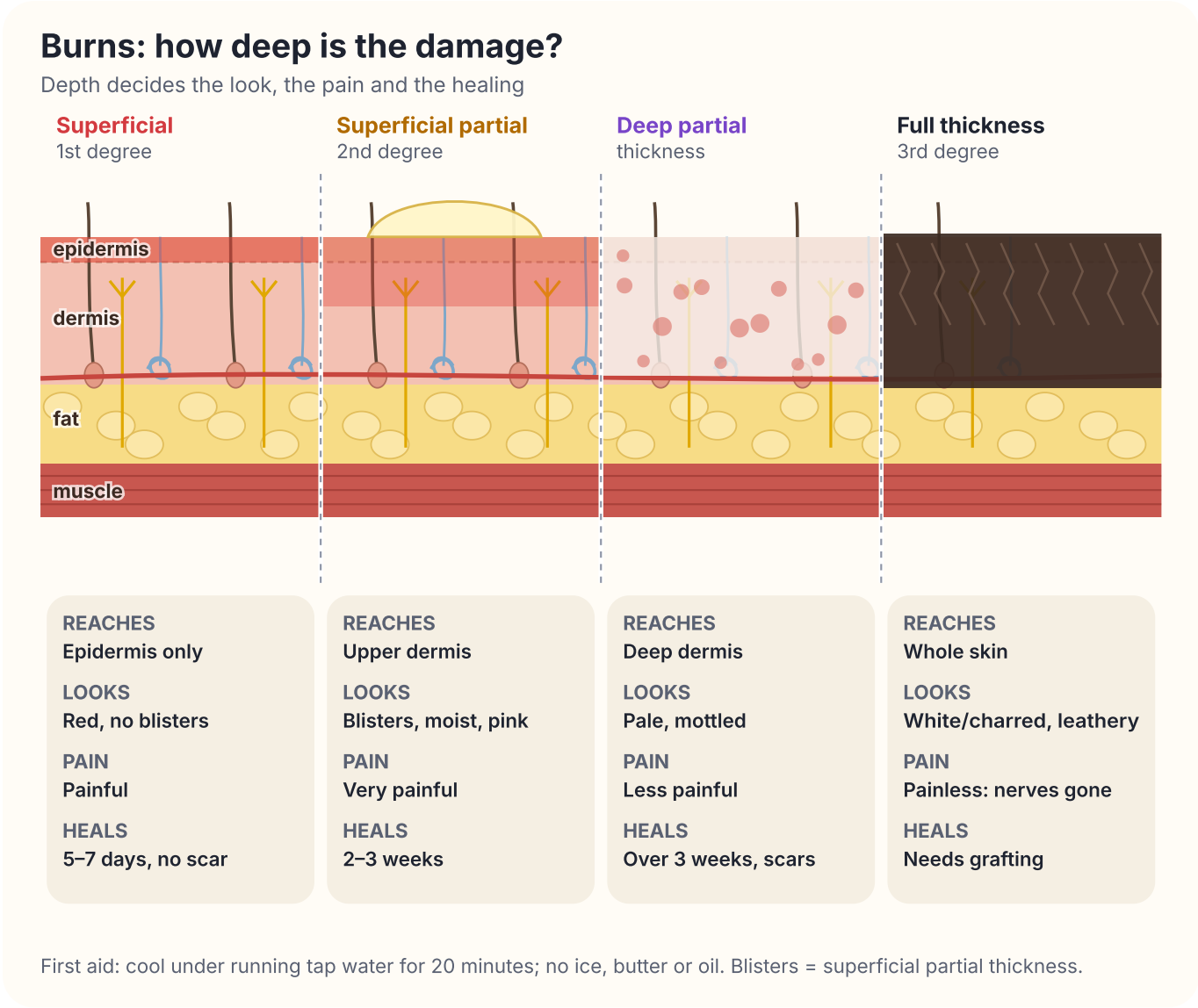
Ulcer	Edge	Typical site
Healing ulcer	Sloping, bluish-pink	—
Tuberculous	**Undermined**	Neck (lymph nodes)
Rodent ulcer (basal cell carcinoma)	**Rolled (raised, beaded)**	Upper face, inner canthus
Squamous cell carcinoma (epithelioma)	**Everted**	Lower lip, old scar (Marjolin's ulcer)
Syphilitic (gumma)	**Punched out**	Leg, palate
Varicose (venous)	Sloping	**Above the medial malleolus (gaiter area)**

Ulcer	Edge	Typical site
Trophic (neuropathic)	Punched out, painless	Heel, ball of the foot (diabetes, leprosy)

Hook: "TB Undermines, Rodent Rolls, Squamous Spills (everted), Syphilis Punches."

Burns

Depth	Layer	Look	Pain	Heals
Superficial (first degree)	Epidermis	Red, no blisters (sunburn)	Painful	5–7 days, no scar
Superficial partial thickness (second degree)	Upper dermis	**Blisters**, moist, pink	Very painful	2–3 weeks
Deep partial thickness	Deep dermis	Pale, mottled	Less painful	Over 3 weeks, scars
Full thickness (third degree)	Whole skin	White or charred, leathery (eschar)	**Painless** (nerves destroyed)	Needs grafting



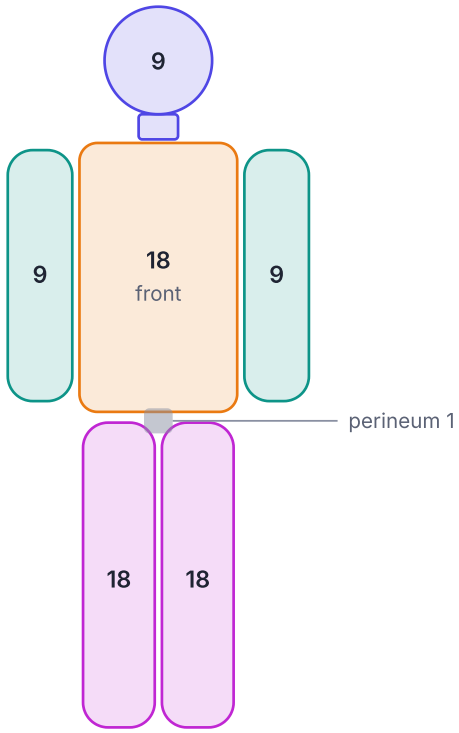
Burn depth · how far down the damage goes decides the look, the pain and the healing

- **Rule of nines (adult)**: head and neck 9%, each upper limb 9%, front of trunk 18%, back of trunk 18%, each lower limb 18%, perineum 1%.
- The patient's own palm (with fingers) is about 1% of body surface.
- In children the head is larger (about 18%) and the legs smaller (Lund and Browder chart).
- Admit adults with over 15% burns (children over 10%), and any burn of the face, hands, feet, perineum or airway.

First aid for burns: stop the burning, **cool with running tap water for 20 minutes** (no ice, butter or oil), remove rings and tight clothing, cover with cling film or a clean cloth, keep the patient warm. Face and neck burns are dangerous because of **airway swelling**.

Rule of nines: each body region is 9% or a multiple of 9%

Adult body, front view; percentages of total body surface area burnt



Region	% of body
Head and neck	9
Each upper limb (×2)	9 + 9
Front of trunk	18
Back of trunk	18
Each lower limb (×2)	18 + 18
Perineum	1
Total	**100**

Patient's palm with fingers = about 1%

Children: head is larger (about 18%) and each leg smaller (about 14%): use the Lund and Browder chart.

Admit adults with burns over 15% (children over 10%).

Rule of nines · adult body surface by region

Add the regions that are burnt: a burn of the whole front of the trunk and one arm is 18 + 9 = 27%.

Perioperative care

- **Before surgery**: informed written consent, fasting (solids 6 hours, clear fluids 2 hours), assessment of fitness (ASA grade I–VI), blood sugar and blood pressure control, marking the site.
- **After surgery**: watch airway, breathing, circulation, pain, urine output; early mobilisation prevents deep-vein thrombosis and chest infection.
- **Post-operative fever**: day 1–2 lung collapse (atelectasis); day 3–5 urinary tract infection or wound infection; day 5–7 deep-vein thrombosis; remember the "5 Ws": wind, water, wound, walking, wonder drugs.

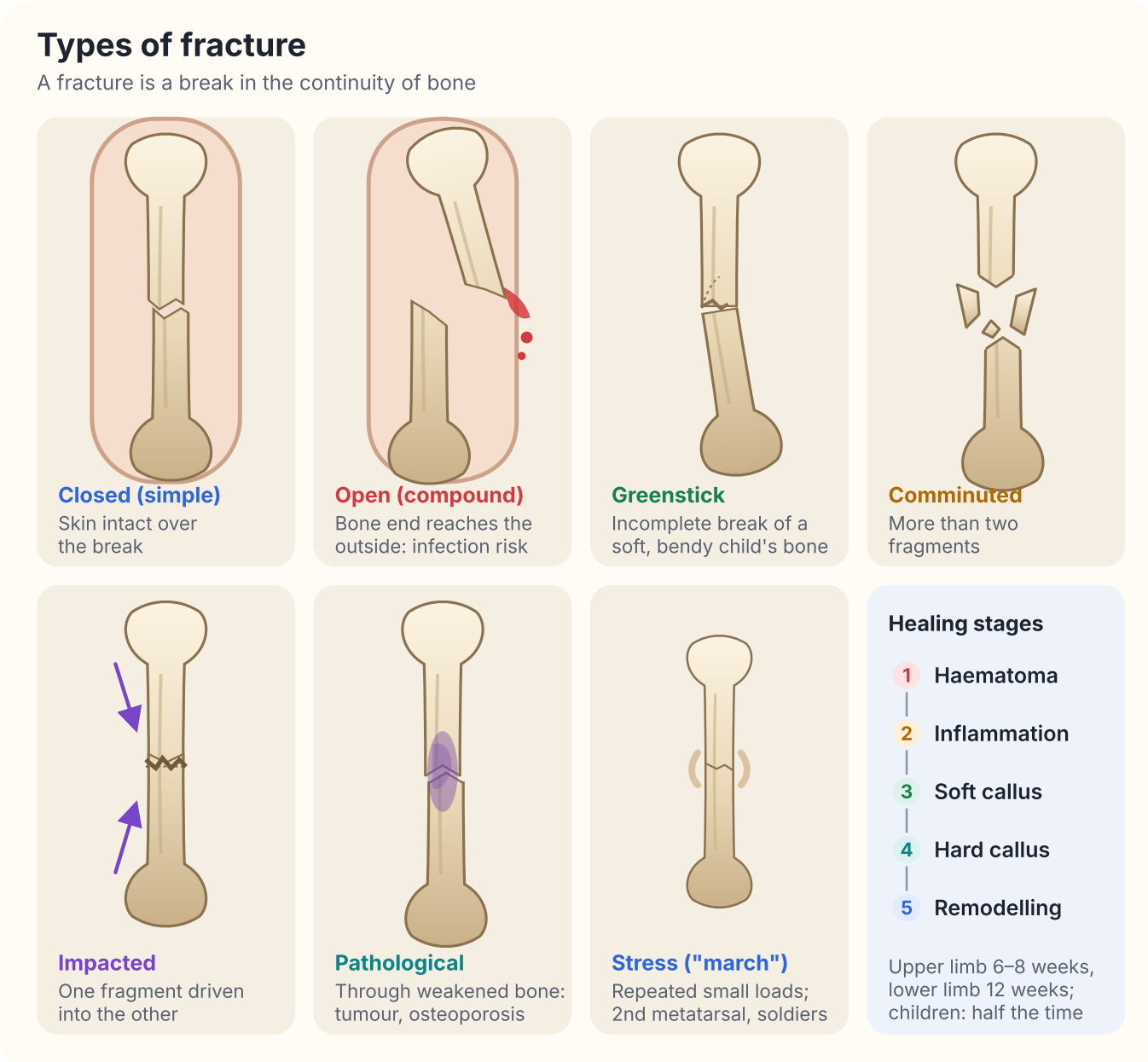
9.2 Traumatology and orthopaedics

Expect eponymous fractures, the difference between a sprain, strain and dislocation, and the X-ray signs of osteoarthritis and spondylosis.

Fractures

A fracture is a break in the continuity of bone. **Signs**: pain, swelling, deformity, abnormal mobility, crepitus, loss of function. **Closed** (skin intact) or **open/compound** (communicates with the outside, risk of infection).

Type	Meaning
Greenstick	Incomplete fracture of a soft, bendy child's bone
Comminuted	More than two fragments
Pathological	Through bone already weakened (tumour, osteoporosis)
Stress (fatigue, "march")	Repeated small loads; 2nd metatarsal in soldiers
Impacted	One fragment driven into the other

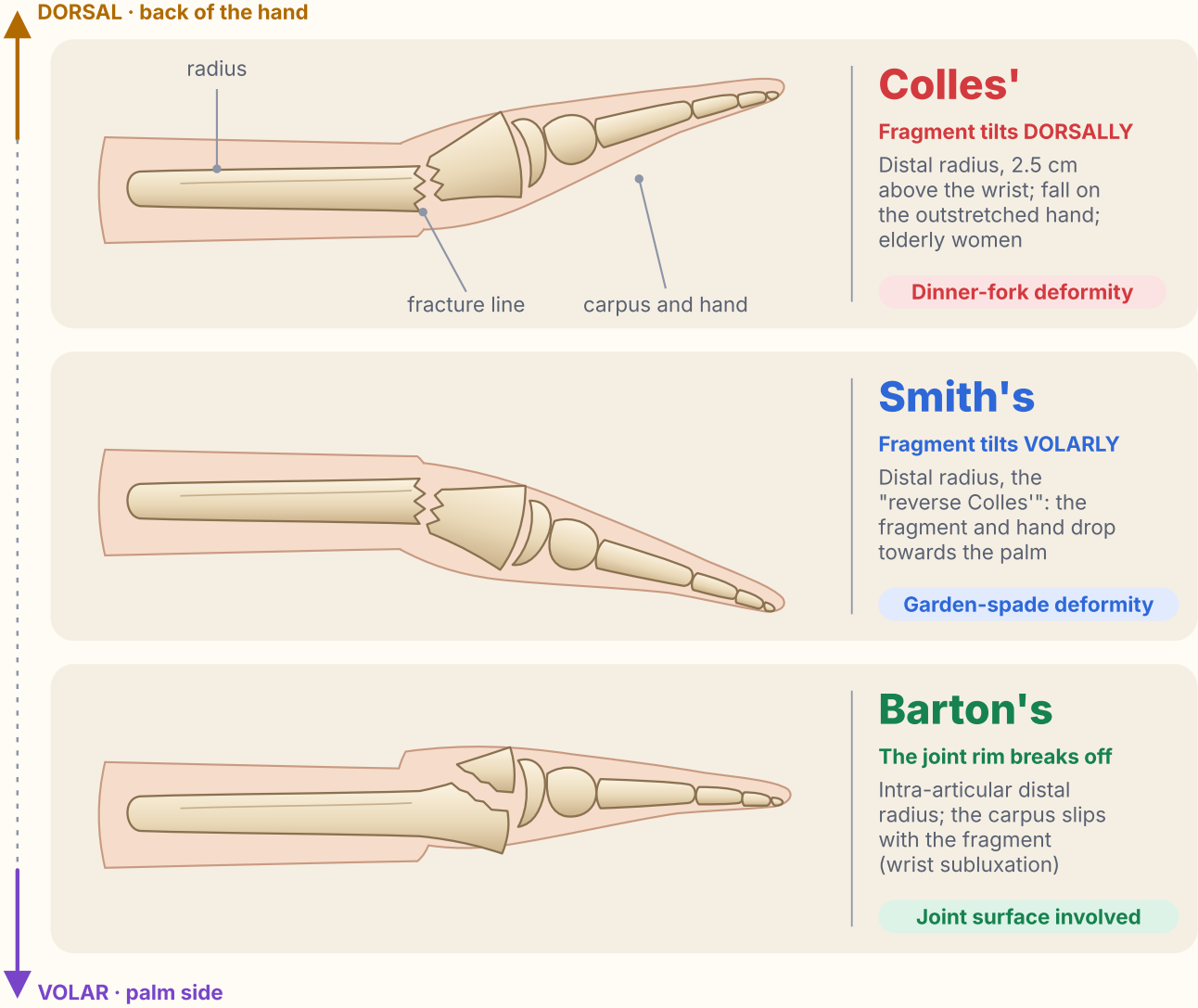


Types of fracture · how each one looks, and the five healing stages

Eponymous fracture	Bone and deformity
Colles'	Distal radius, about 2.5 cm above the wrist, **dorsal displacement** ("dinner-fork" deformity); fall on the outstretched hand, elderly women
Smith's	Distal radius, **volar** displacement ("reverse Colles'", garden-spade deformity)
Barton's	Intra-articular distal radius with subluxation of the wrist
Monteggia	Fracture of upper ulna + dislocation of the radial head (**MU**: Monteggia = Ulna)
Galeazzi	Fracture of lower radius + dislocation of the distal radio-ulnar joint
Pott's	Bimalleolar fracture of the ankle
Bennett's	Fracture-dislocation of the base of the first metacarpal (thumb)
Boxer's	Neck of the 5th metacarpal
Jones	Base of the 5th metatarsal

Wrist fractures: Colles', Smith's and Barton's

Same bone, same place, different direction · side view of the right forearm



Wrist fractures · Colles' tilts dorsally, Smith's volarly, Barton's breaks the joint rim

- **Healing stages**: haematoma → inflammation → soft callus → hard callus → remodelling.
- Upper limb bones unite in about 6–8 weeks, lower limb in 12 weeks; children heal in half the time.
- **Complications**: shock, fat embolism (long bones, 24–72 h), compartment syndrome (pain on passive stretch: the earliest sign), avascular necrosis (scaphoid, femoral head), malunion, non-union, Volkmann's ischaemic contracture (supracondylar fracture of humerus in children).

First aid: immobilise the joint above and below with a splint, do not try to straighten deformity, cover open wounds, treat for shock.

Dislocations and soft-tissue injury

Term	Meaning
Dislocation	Complete loss of contact between joint surfaces

Term	Meaning
Subluxation	Partial loss of contact
Sprain	Injury to a **ligament** (commonest: lateral ligament of the ankle, inversion injury)
Strain	Injury to a **muscle or tendon**

Shoulder is the commonest joint to dislocate, **anterior** in about 95% (axillary nerve at risk). **Hip** dislocation is usually **posterior** (dashboard injury; limb flexed, adducted, internally rotated; sciatic nerve at risk). **RICE** for sprains: Rest, Ice, Compression, Elevation.

Osteomyelitis

- Infection of bone.
- Acute haematogenous osteomyelitis is commonest in children, in the **metaphysis** of long bones (lower femur, upper tibia), caused mostly by ***Staphylococcus aureus***.
- Signs: high fever, severe local pain, refusal to move the limb.
- X-ray is normal for the first 10–14 days; MRI shows it early.
- Chronic osteomyelitis: **sequestrum** (dead bone), **involucrum** (new bone around it), discharging sinus.
- Tuberculosis of the spine is **Pott's disease** (gibbus, cold abscess, paraplegia).

Degenerative joint disease

	Osteoarthritis	Rheumatoid arthritis (for contrast)
Age	Over 50, weight-bearing joints	Young and middle-aged women
Joints	Knee, hip, DIP (**Heberden's nodes**), PIP (**Bouchard's nodes**)	Small joints of hands, symmetrical, MCP and PIP
Stiffness	Under 30 minutes, worse after use	Morning stiffness over 1 hour
X-ray	**LOSS**: Loss of joint space, Osteophytes, Subchondral sclerosis, Subchondral cysts	Erosions, osteopenia

- **Cervical spondylosis**: neck pain and stiffness radiating to the shoulders and arms, numbness and tingling (radiculopathy, commonest at C5–C6); vertigo on turning the head.
- **Lumbar spondylosis / disc prolapse**: low back pain with sciatica (L4–L5 and L5–S1), positive straight-leg-raising test.

- **Frozen shoulder** (adhesive capsulitis): painful restriction of all shoulder movements, commoner in diabetics.

Homoeopathic therapeutics

Picture	Remedy
Sprain or strain; stiffness worse on first motion, better from continued motion	**Rhus toxicodendron**
Bruised tendons and periosteum, sprained wrists and ankles, eye strain	**Ruta graveolens**
Fractures slow to unite; pricking pain at the site	**Symphytum**
Delayed union in a thin, anaemic patient	**Calcarea phosphorica**
Osteoarthritis with joint pain worse from the slightest motion, better from pressure and rest	**Bryonia alba**
Cervical spondylosis with stiffness and vertigo, worse on turning the head	**Conium**, Cocculus
Sciatica with cramping pain, better from pressure and heat; left side	**Colocynthis**
Sciatica with numbness, worse sitting, better lying	**Gnaphalium**
Osteomyelitis with chronic suppuration, fistulous openings, offensive discharge	**Silicea**; bone necrosis: **Calcarea fluorica**

Memory hook: the **shoulder** slips **forward** (anterior), the **hip** is pushed **back** (posterior, dashboard injury). In osteomyelitis the **S**equestrum is the **S**lain bone and the **I**nvolucrum is the new bone that **I**nvests it.

9.3 ENT and ophthalmology

Six conditions are named in the syllabus; for each, know the classic picture, one complication and one or two remedies.

ENT

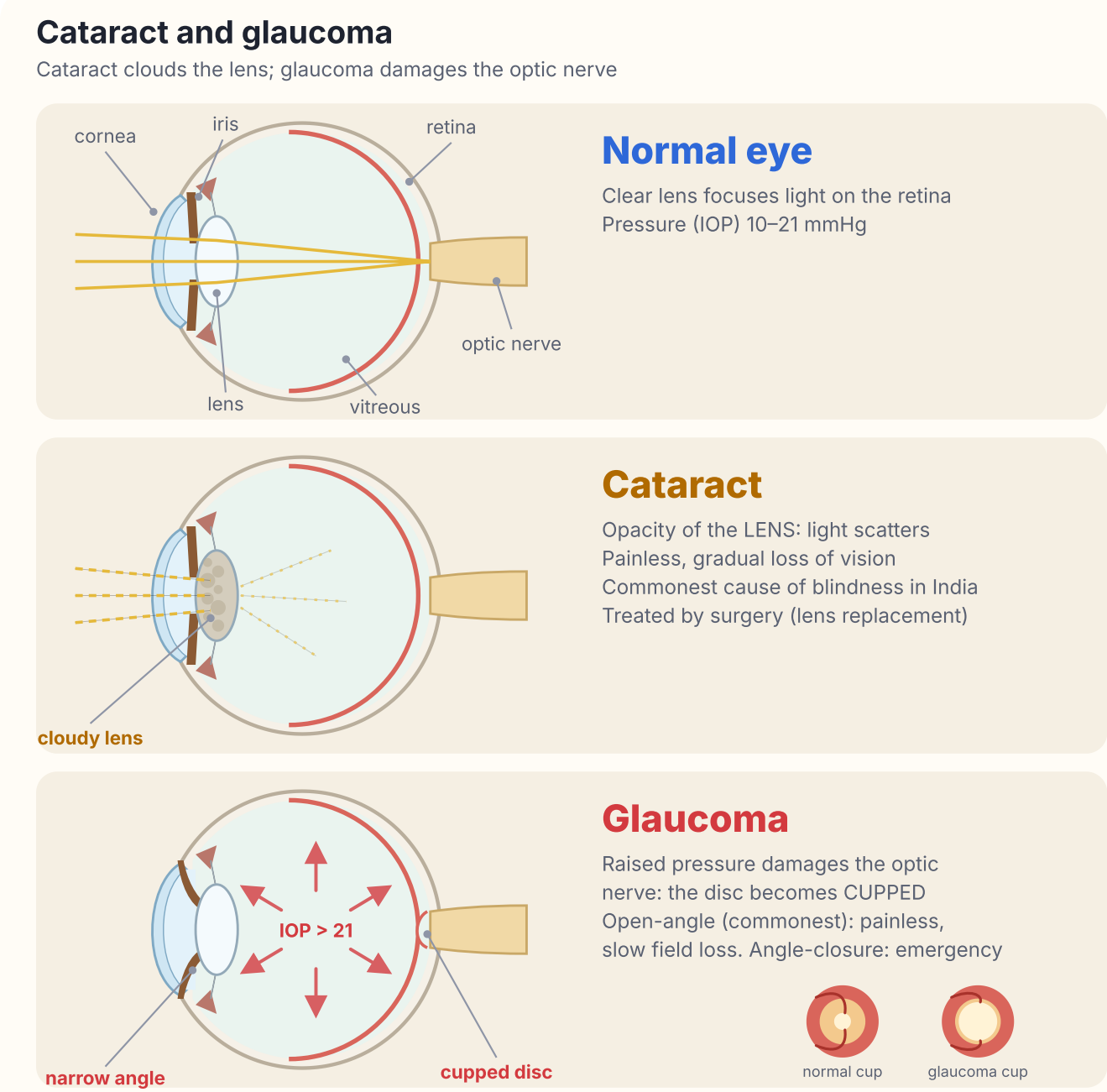
Condition	Key facts	Remedies
Sinusitis	Commonest sinus affected in adults: **maxillary** (drains poorly, opening high up); in children: ethmoid. Facial pain worse on bending forward, nasal block, purulent discharge, post-nasal drip	Kali bichromicum (thick, ropy, stringy discharge; pain at the root of the nose); Hepar sulph; Silicea (chronic)
Acute otitis media	Middle-ear infection after a cold, commonest in children (short, horizontal Eustachian tube). Earache, fever, **red, bulging drum**; if it bursts, discharge and pain relief. Chronic suppurative otitis media (CSOM): safe (tubotympanic, central perforation) or unsafe (atticoantral, cholesteatoma)	Belladonna (sudden, throbbing, red), Chamomilla (unbearable pain, angry child), Pulsatilla (bland discharge, worse warm room), Ferrum phos (first stage)
Tonsillitis	Sore throat, fever, painful swallowing, enlarged red tonsils with exudate; commonest bacterial cause **Group A beta-haemolytic streptococcus** (risk of rheumatic fever and glomerulonephritis). Peritonsillar abscess = **quinsy**	Belladonna (right side, bright red), Lachesis (left to right, worse empty swallowing, hot drinks), Lycopodium (right to left), Baryta carb (recurrent, enlarged tonsils), Hepar sulph (suppuration)
Otosclerosis (contrast)	Progressive conductive deafness in young women, worse in pregnancy; hears better in noise (paracusis Willisii)	—
Epistaxis	Commonest site: **Little's area** (Kiesselbach's plexus, anteroinferior septum); first aid: sit up, lean forward, pinch the soft nose for 10 minutes	Ferrum phos, Millefolium, Phosphorus

Hearing tests. Rinne: air conduction better than bone = positive (normal or sensorineural loss); negative in conductive loss. Weber: lateralises to the **affected** ear in conductive loss, to the **better** ear in sensorineural loss.

Eye

Condition	Key facts	Remedies
Conjunctivitis	Red eye with discharge, vision normal, pupil normal. Bacterial: sticky purulent; viral: watery, often with preauricular nodes; allergic: itching, ropy discharge (spring catarrh in boys)	Euphrasia (acrid tears, bland nasal discharge), Argentum nitricum (profuse purulent, ophthalmia neonatorum), Pulsatilla (bland yellow discharge), Apis (swollen lids, stinging)

Condition	Key facts	Remedies
Cataract	Opacity of the lens; painless, gradual loss of vision; **commonest cause of blindness in India**; senile cataract commonest; treated by surgery (lens replacement)	Calcarea fluorica, Causticum, Phosphorus, Cineraria maritima (local) in early cases
Glaucoma	Raised intraocular pressure (normal **10–21 mmHg**) damaging the optic nerve (cupping); **acute angle-closure**: sudden painful red eye, **halos round lights**, vomiting, fixed semi-dilated oval pupil — an emergency. **Open-angle** (commonest): painless, slow field loss	Belladonna, Phosphorus, Spigelia (pain in and around the eye), Osmium
Stye (hordeolum)	Painful infection of a lash follicle at the lid margin	Pulsatilla, Staphysagria (recurrent)
Chalazion	Painless swelling of a meibomian gland, away from the margin	Staphysagria, Thuja



Cataract and glaucoma · the lens clouds in one, the optic nerve cups in the other

Leukocoria (white pupil) in a child must raise suspicion of **retinoblastoma**. Vitamin A deficiency causes night blindness and Bitot's spots. A chemical splash: wash the eye at once with plenty of clean water for 15–30 minutes.

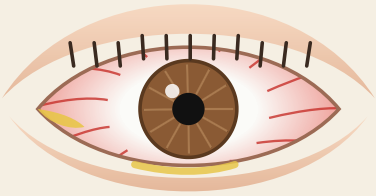
Red eye decision chart: check pain, vision, pupil and pressure to tell the three common red eyes apart.

Feature	Acute conjunctivitis	Acute angle-closure glaucoma	Acute iritis (anterior uveitis)
Pain	Grittiness, little pain	Severe, with headache and vomiting	Moderate ache, photophobia
Vision	Normal	Markedly reduced, haloes round lights	Mildly blurred

Feature	Acute conjunctivitis	Acute angle-closure glaucoma	Acute iritis (anterior uveitis)
Redness	Conjunctival, worst at fornices	Ciliary flush	Ciliary (circumcorneal) flush
Discharge	Present (mucopurulent or watery)	None, watering only	None, watering only
Cornea	Clear	Hazy, oedematous	Clear; keratic precipitates
Pupil	Normal, reacting	Mid-dilated, vertically oval, fixed	Small, irregular, sluggish
Intraocular pressure	Normal	Very high (stony hard eye)	Normal or low
Key action	Hygiene, treat at home	**Emergency referral** to save sight	Refer for slit lamp examination
Leading remedies	Euphrasia, Argentum nitricum, Pulsatilla	Spigelia, Phosphorus, Belladonna (only alongside urgent referral)	Mercurius corrosivus, Rhus toxicodendron

Three red eyes: look at the pupil

A red eye with a fixed mid-dilated pupil is glaucoma until proved otherwise



Acute conjunctivitis

PAIN
Gritty, little pain

VISION
Normal

REDNESS
Worst at the fornices

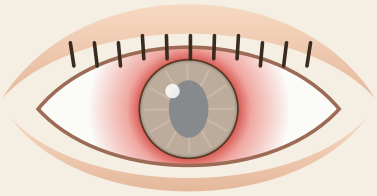
DISCHARGE
Present

CORNEA
Clear

PUPIL
Normal, reacting

PRESSURE
Normal

Hygiene, treat at home



Acute angle-closure glaucoma

PAIN
Severe, headache, vomiting

VISION
Much reduced, halos

REDNESS
Ciliary flush

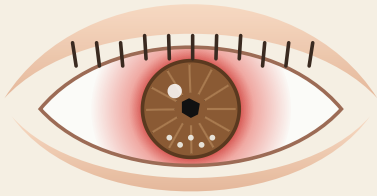
DISCHARGE
None, watering

CORNEA
Hazy, oedematous

PUPIL
Mid-dilated, oval, fixed

PRESSURE
Very high, stony hard

EMERGENCY referral



Acute iritis
(anterior uveitis)

PAIN
Moderate ache, photophobia

VISION
Mildly blurred

REDNESS
Ciliary flush

DISCHARGE
None, watering

CORNEA
Clear; keratic precipitates

PUPIL
Small, irregular, sluggish

PRESSURE
Normal or low

Refer for slit lamp

Three red eyes · the pupil and the pressure tell them apart

☆ **Exam tip:** a red eye with a **fixed mid-dilated pupil** is glaucoma until proved otherwise; a red eye with a **small pupil** points to iritis.

Memory hook: Weber **comes to** a conductive loss and **runs from** a nerve loss: the sound goes to the **affected** ear in conductive deafness and to the **better** ear in sensorineural deafness.

9.4 Surgical therapeutics and first aid

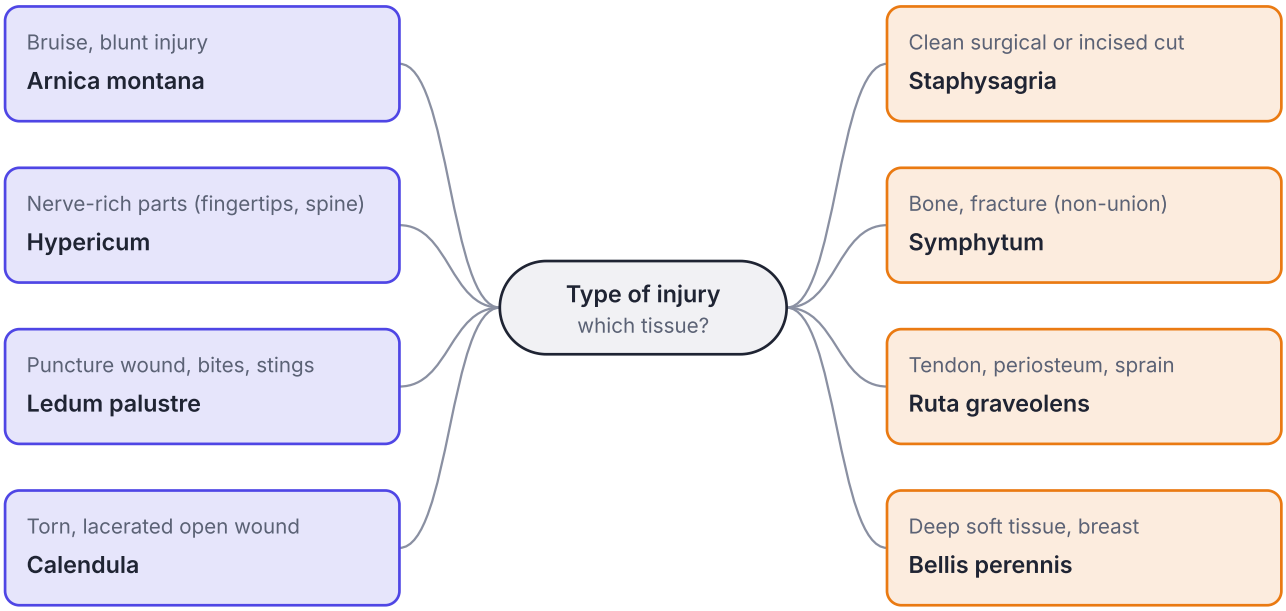
The injury remedies are among the most reliable marks in the paper: each remedy is matched to one kind of tissue or wound.

Remedies for mechanical injuries

The map below pairs each type of injury with its remedy; the table adds the keynote.

Pick the remedy by the tissue injured

Eight injury remedies; first aid and surgical referral come first



Injury → remedy map · remedy chosen by the tissue injured

Match the tissue first: a nail puncture is Ledum, a crushed fingertip is Hypericum, and a fracture slow to unite is Symphytum.

Injury	Remedy	Keynote
Blunt injury, bruises, shock of injury, concussion	**Arnica montana**	Sore, bruised feeling; says he is well; bed feels too hard
Injury to nerve-rich parts (fingertips, spine, coccyx); shooting pains along nerves	**Hypericum**	"The Arnica of the nerves"; also prevents tetanus-like pains after punctured nerve injuries
Punctured wounds, insect bites, rat bites; wound cold yet better from cold	**Ledum palustre**	Ascending pains; prevents sepsis in punctured wounds
Lacerated, open wounds; promotes healing and prevents suppuration	**Calendula** (local and internal)	"The homoeopathic antiseptic"
Clean incised (surgical) wounds; pain after operations and catheterisation	**Staphysagria**	Sense of insult or humiliation
Fractures slow to unite; injury to the eyeball by a blunt object	**Symphytum**	"Knitbone"

Injury	Remedy	Keynote
Bruised tendons, periosteum and joints; eye strain	**Ruta graveolens**	Lameness after sprain
Deep bruising of soft tissues and breasts; injury of deep pelvic organs	**Bellis perennis**	Bruised soreness of deep tissues after surgery
Haemorrhage, bright red, from injury	**Millefolium**, Arnica	—
Burns with blisters and burning pain	**Cantharis** (also Urtica urens for first-degree)	Better from cold applications
Ill effects of a fall on the head, concussion	Arnica, **Natrum sulphuricum** (mental symptoms after head injury)	—

Hook: "ARNICA for the Bruise, HYPERICUM for the Nerve, LEDUM for the Needle, CALENDULA for the Cut-and-Torn, STAPHYSAGRIA for the Scalpel, SYMPHYTUM for the Skeleton, RUTA for the Rope (tendon)."

Pre- and post-operative remedies

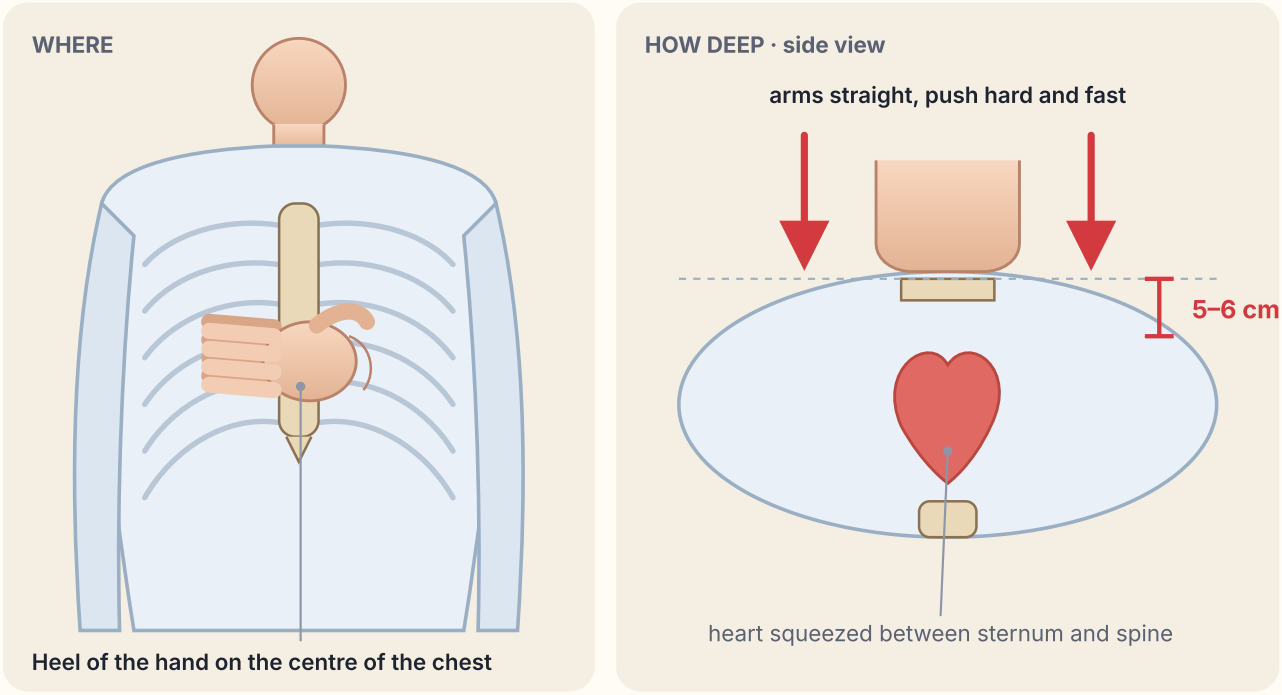
Situation	Remedy
Fear and anticipatory anxiety before surgery, trembling, diarrhoea	**Gelsemium**, Argentum nitricum
Restless fear of death before surgery	**Aconitum napellus**
Shock, bruised soreness after surgery	**Arnica**
Pain of a clean incision	**Staphysagria**
Retention of urine after surgery	**Causticum**
Flatulent distension after abdominal surgery	**Carbo vegetabilis**, Raphanus
Vomiting after anaesthesia	**Phosphorus**
Ill effects of anaesthesia, slow recovery, sleepiness	**Opium**
Collapse with cold breath and craving for air	**Carbo vegetabilis**

First aid

Basic life support (adult) — C-A-B. Check response and breathing → call for help → chest compressions **30 : 2 breaths**, rate **100–120 per minute**, depth 5–6 cm, centre of the chest; continue until help arrives. Recovery position for an unconscious patient who is breathing.

CPR: 30 compressions, 2 breaths

Where to push, how deep and how fast



Adult basic life support: C-A-B (Circulation, Airway, Breathing)

Check response and breathing → **Call** for help → **30 compressions** 100–120 per minute → **2 breaths** then repeat 30 : 2

Breathing but unconscious: recovery position. Continue CPR until help arrives.

CPR · centre of the chest, 5–6 cm deep, 100–120 a minute, 30 : 2

Emergency	First aid
Snakebite	Reassure, **immobilise the limb** like a fracture, remove rings and tight items, keep the patient still, rush to hospital for anti-snake venom. **Do not** cut, suck, apply a tight tourniquet, ice or herbs ("Do it RIGHT": Reassure, Immobilise, Get to hospital, Tell the doctor the symptoms)
Neurotoxic bites (cobra, krait)	Ptosis, difficulty swallowing, respiratory paralysis; krait bites at night while asleep, often painless

Emergency	First aid
Haemotoxic bites (Russell's viper, saw-scaled viper)	Local swelling, bleeding from gums and bite site, kidney failure; 20-minute whole-blood clotting test (20WBCT)
Poisoning (general)	Remove from the source, clear the airway, place in the recovery position, keep the container or sample, do not induce vomiting after corrosives or kerosene, shift to hospital
Corrosive or kerosene swallowed	No vomiting (risk of further burns or aspiration pneumonia)
Bleeding	Direct pressure and elevation; tourniquet only as a last resort for life-threatening limb bleeding
Fracture	Immobilise the joints above and below; do not straighten
Choking (adult)	5 back blows, then 5 abdominal thrusts (Heimlich)
Heat stroke	Move to shade, remove clothing, cool with water and fanning
Drowning	Clear airway, start CPR with rescue breaths
Dog bite	Wash the wound with soap and running water for 15 minutes; rabies vaccination

Homoeopathic support in bites and stings. Ledum (punctured bites), Apis (stinging, oedematous swelling), Lachesis (bluish, septic bite wounds), Arsenicum album (collapse, anxiety, burning). Snakebite is always treated with anti-snake venom in hospital; remedies are supportive only.

Rapid revision: one-line facts

Area	Fact
Wounds	First intention = clean incised wound, minimal scar; second intention = granulation, wide scar
Wounds	Keloid grows beyond the margins; hypertrophic scar stays within
Sinus / fistula	Sinus = blind track; fistula = link between two epithelial surfaces
Infection	Abscess = Staph. aureus, drain it; cellulitis = Strep. pyogenes, spreading
Shock	Commonest after trauma = hypovolaemic; neurogenic = low BP with slow pulse

Area	Fact
Ulcers	TB undermined; rodent rolled; squamous everted; syphilitic punched out; varicose above the medial malleolus
Burns	Blisters = superficial partial thickness; painless = full thickness; cool 20 minutes under running water
Rule of nines	Head 9, each arm 9, front trunk 18, back trunk 18, each leg 18, perineum 1
Fractures	Colles' = distal radius, dorsal displacement, dinner fork; Smith's = volar
Fractures	Monteggia = ulna + radial head; Galeazzi = radius + distal radio-ulnar joint; Pott's = ankle
Fractures	Compartment syndrome earliest sign = pain on passive stretch; scaphoid and femoral neck = avascular necrosis
Joints	Sprain = ligament; strain = muscle or tendon; shoulder dislocates anteriorly, hip posteriorly
Bone	Osteomyelitis = metaphysis, Staph. aureus; sequestrum = dead bone; involucrum = new bone
Osteoarthritis	Heberden's nodes (DIP), Bouchard's (PIP); X-ray LOSS
ENT	Maxillary sinus commonest; red bulging drum = acute otitis media; quinsy = peritonsillar abscess; Little's area = nosebleed
Eye	Cataract = lens opacity, commonest cause of blindness in India; glaucoma IOP over 21 mmHg, halos = acute angle closure
Eye	Leukocoria in a child = retinoblastoma until proven otherwise
Remedies	Arnica bruise; Hypericum nerve; Ledum puncture; Calendula torn wound; Staphysagria incision; Symphytum fracture; Ruta tendon; Bellis deep tissue
Remedies	Pre-op fear = Gelsemium, Aconite; post-op retention = Causticum; vomiting after anaesthesia = Phosphorus; flatulence = Carbo veg
First aid	CPR 30 : 2, 100–120/min, 5–6 cm; snakebite: immobilise, no cutting or sucking, hospital for ASV

One-glance recall card

Topic 9 recall card: the 15 facts asked most

Cover the small lines, say the fact from the big cue, then check

Rule of nines head 9, arm 9, trunk 18+18, leg 18, perineum 1	**Palm = 1%** patient's palm with fingers is about 1% of the body	**Cool water** burn first aid: 20 min, no ice, no butter
30 : 2 CPR, 100-120 a minute, 5-6 cm deep	**Shoulder: anterior** commonest dislocation; hip goes posterior	**Dinner fork** Colles' fracture of the distal radius
Metaphysis osteomyelitis site; Staph aureus	**Sequestrum** dead bone; involucrum is the new bone around it	**Weber** to the affected ear: conductive; to the better ear: nerve
Fixed mid-dilated red eye with this pupil is acute glaucoma	**Leukocoria** white pupil in a child: think retinoblastoma	**Arnica** bruises, blunt injury, shock of injury
Ledum punctures and stings; cold yet better cold	**Hypericum** nerve-rich parts: fingertips, spine	**Symphytum** fracture slow to unite; Staphisagria: clean cuts

Topic 9 recall card · 15 tiles

For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

Practice MCQs

Twenty questions grouped by syllabus heading; the answer key follows.

General surgical conditions

1. An abnormal communication between two epithelial surfaces is a: (a) sinus (b) fistula (c) abscess (d) ulcer
2. An ulcer with an undermined edge is typical of: (a) basal cell carcinoma (b) tuberculosis (c) syphilis (d) varicose veins
3. Burns with blisters and severe pain are: (a) superficial (first degree) (b) superficial partial thickness (c) full thickness (d) fourth degree
4. By the rule of nines, one entire lower limb of an adult is: (a) 9% (b) 18% (c) 27% (d) 36%

5. Low blood pressure with a slow pulse after spinal injury suggests: (a) hypovolaemic shock (b) neurogenic shock (c) septic shock (d) cardiogenic shock

Trauma and orthopaedics

6. Colles' fracture involves the distal radius with: (a) volar displacement (b) dorsal displacement (c) ulnar dislocation (d) scaphoid fracture
7. A sprain is an injury to a: (a) bone (b) muscle (c) ligament (d) nerve
8. Acute haematogenous osteomyelitis in children most commonly affects the: (a) epiphysis (b) metaphysis (c) diaphysis (d) flat bones
9. Heberden's nodes in osteoarthritis are found at the: (a) MCP joints (b) PIP joints (c) DIP joints (d) wrist
10. The commonest direction of shoulder dislocation is: (a) anterior (b) posterior (c) inferior (d) superior

ENT and eye

11. Peritonsillar abscess is called: (a) quinsy (b) Ludwig's angina (c) adenoiditis (d) laryngitis
12. The commonest site of nosebleed is: (a) Woodruff's area (b) Little's area (c) the turbinates (d) the sphenoid sinus
13. Sudden painful red eye with halos around lights and vomiting suggests: (a) conjunctivitis (b) acute angle-closure glaucoma (c) cataract (d) stye
14. Opacity of the lens of the eye is: (a) glaucoma (b) cataract (c) pterygium (d) keratitis
15. Tonsillitis beginning on the left and moving to the right, worse from hot drinks: (a) Lycopodium (b) Lachesis (c) Belladonna (d) Baryta carb

Surgical therapeutics and first aid

16. Injury to nerve-rich parts such as crushed fingertips calls for: (a) Arnica (b) Hypericum (c) Calendula (d) Ruta
17. Punctured wounds that are cold yet better from cold applications: (a) Ledum (b) Arnica (c) Staphysagria (d) Symphytum
18. Pain after a clean surgical incision with a sense of insult: (a) Bellis perennis (b) Staphysagria (c) Hypericum (d) Ruta
19. First aid for snakebite includes: (a) cutting and sucking the wound (b) a tight tourniquet (c) immobilising the limb and urgent transfer to hospital (d) ice packs
20. In adult CPR the ratio of chest compressions to breaths is: (a) 15 : 2 (b) 30 : 2 (c) 5 : 1 (d) 30 : 1

Answer key

Q	Answer	Why
1	b	Sinus is blind; fistula links two surfaces
2	b	TB = undermined edge
3	b	Blisters + pain = superficial partial thickness
4	b	Each lower limb 18%
5	b	Loss of sympathetic tone: low BP, slow pulse
6	b	Dinner-fork deformity
7	c	Strain = muscle or tendon
8	b	Sluggish metaphyseal blood flow
9	c	Heberden = DIP; Bouchard = PIP
10	a	About 95% anterior
11	a	Quinsy = peritonsillar abscess
12	b	Kiesselbach's plexus on the anterior septum
13	b	Emergency: raised IOP
14	b	Cataract
15	b	Lachesis: left to right, worse hot drinks
16	b	Hypericum = nerve injuries
17	a	Ledum keynote
18	b	Staphysagria = incised wounds
19	c	No cutting, sucking or tight tourniquet
20	b	30 : 2 at 100–120/min